Date: September 20, 1961

From (Dept/Loc): Research Center
Yorktown Hgts., N.Y.

IBM

Subject and/or Reference: Super-Stretch Memos

To: H. G. Kolsky

Enclosed are copies of the memos on potential Super-Stretch applications prepared by our group last Spring.

No unified report was ever undertaken.

Bill

W. R. Elmendorf

WRE:eaw
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Machine Organization
Technical Memo #20

YORKTOWN
RESEARCH

February 1, 1961

A Sketch of Weather Data Processing

by

L. R. Johnson

1. Introduction

The unclassified sector of the U. S. weather information complex is the subject of this report. The complex uses data processing resources in two ways: (a) to help in the operation of a weather information system, and (b) to help in enabling system improvements. The purpose of this report is to provide an introduction to the whole complex, with particular emphasis on present or planned requirements for automatic data processing.

2. Structure of Weather Science

2.1 Observation

Weather phenomena of general interest have traditionally included temperature, pressure, humidity, wind velocity, visibility, cloud cover, cloud type, precipitation amount and precipitation type. For purposes of quantifying a climate or of predicting near-future weather, there is little information in an isolated observation. To generate useful meteorological information, it is necessary to sample over both space and time.

The number of sampling points, the geometry of the points, and the sampling rate are matters which must be decided on the basis of available theory, available resources, and operational needs. For climatological uses it is thought desirable to employ a high density of surface observations, particularly over land masses. In providing analytical information for forecasting purposes, however, the direction for years has been towards more extensive networks and increased reliance upon upper-air data. U. S. Weather Bureau forecasters now use a peace-time grid of upper-air soundings that samples the whole Northern Hemisphere. Forecasts of motion in the upper atmosphere are not only of growing importance in themselves -- i. e., in the aviation, ballistics and air pollution areas -- but are widely used to help derive forecasts for surface weather conditions. Because of growing attention to forecasts of an hour or two, a counter trend towards dependence on low-level observations for short-range forecasts also exists.

Considerable data processing separates raw observational data and the compact analytical charts or summaries needed for prediction purposes. Data must be converted to digital form, reduced to standard conditions and often to derived quantities, put into message form, and transmitted to a message center. Messages are assembled, further analyzed as a group, and various display maps are plotted and drawn. The analytical displays are then transmitted to local forecasting centers.

2.2 Subjective forecasting

Because air is heated in lower and cooled in higher latitudes, the earth's atmosphere is a heat engine. The massive energy exchange maintains certain fluid flows that underly global weather processes. Variations in the earth's surface and in its radiation balances lead to all kinds of perturbations in the flow, however. Meteorologists have a somewhat arbitrary scheme for classifying perturbations or changes. The biggest changes are those that alter the general circulation pattern and are brought about partly by seasonal variations, partly by combinations of less-predictable events. Superimposed on the general circulation pattern are big storms, little storms, and minute events.

Modern forecasting really emerged after serious attention was given to the fact that weather, particularly in the temperate zones, is strongly influenced by distinctive air masses and by their boundary clashes as they are towed along in the general circulation pattern. From analyses of air mass characteristics, pressure gradients and dominant features of the circulation at selective levels in the atmosphere, the forecaster assimilates a feel for the whole weather picture. Applying his knowledge and experience, he selects certain features as most relevant and analyzes these more carefully. Then he (actually different steps may be done by different people) makes three prognoses in turn:

(1) a forecast of circulation in the middle atmosphere, e.g., a map of the altitude of the 500 millibar surface;

(2) a forecast of the position of air masses and their boundaries in the lower atmosphere; and

(3) a forecast of weather phenomena on the ground or over airways, as desired.

Forecast periods may range from a few hours to three days. Where more responsive forecasts are needed, as for severe storms or airport conditions, simpler and more direct techniques have generally been used. Forecasts for more than three days are also likely to use special techniques.

Until recently, all three forecasts would have been made subjectively, but advances in theory have reached the point where forecast (1) can be routinely accomplished on a computer. A responsible forecaster may modify or reject the objective forecast, but in fact this is seldom done, and the process has become essentially objective in the U. S. Weather Bureau.

2.3 Objective forecasting

Subjective weather forecasting has three main shortcomings. First, the training and maintenance of forecasters is expensive. Second, it takes significant amounts of time to summarize information for a forecaster, for him to assimilate the information, and for him to document his forecast. Third, subjective forecasting is practically a static art; any performance jumps would seem to depend upon a much more thorough analysis of interrelationships than is possible for a human being.

The two main approaches to objective forecasting differ in their choices of mathematical models. One approach assumes that meteorological time series are stationary, over some period of interest, and uses a regression model to extrapolate a series. There are clearly a number of possibilities here, including (a) autocorrelation, (b) multiple regression with lagged dependent variables, and (c) multiple regression with forecasted dependent variables. Possibility (b) is sometimes mentioned as a tool for predicting movements in the general circulation pattern, hence for long-range forecasts, but this has not met with any great success. The use of (b) for short-range,limited-area forecasts is suggestive. The use of possibility (c) in step (3) of the forecasting cycle to quantify forecasts of precipitation, visibility, ceiling height, cloud cover, etc., is also thought to hold some promise.

The second approach to objective forecasting sets out to approximate the whole heat engine by a dynamic model simplified out of the basic laws of mechanics, of thermodynamics, of gases, and of radiation. The designer of a dynamic model is faced with formidable problems. He cannot step through controlled experiments; he must make the best of very imperfect data; he must operate on the frontiers of numerical analysis; and he must deal, at the minimum, with a fairly large mass of data. Progress is being made, nevertheless, and several theorists are actively pushing the dynamic approach. A dynamic model is used to produce the 500 millibar objective forecast mentioned above. An up-to-date technical survey of the dynamic model approach has been prepared by Phillips [1].

2.4 Directions

The U. S. Weather Bureau has a wide variety of research interests, but only two of these have much potential for altering both weather theory

and practice. The first of these is a study of dynamic models carried on in the General Circulation Research Section. A technical paper on the computational interests of this project has been prepared by Smagorinsky [2]. A recent Weather Bureau document [3, p. 48] comments on Smagorinsky's "Mark II" model as follows: "The complete hydrodynamic system of equations including lateral diffusion, the method of mapping, the vertical structuring, the condensation mechanism, the radiation transfer mechanism, the boundary layer approximations, and the convective transfer mechanism have been worked out and the problem is being programmed for a computer." Success in this work could lead to improved objective forecasting on earth, to forecasts of atmospheric behavior for other planets, and to a theoretical assessment of proposed methods for modifying weather and climate.

The other interesting development lies in the possibility of obtaining better data through the use of weather satellites. Information on the radiation balance over the earth has always been inadequate. Radiation measurements in Explorer VII (launched 13 October 1959) dramatized the possibility for new weather data. Photographs from Tiros I (launched 1 April 1960) also demonstrated a data source that could help pinpoint the position and development of large and small storms. The question of photograph interpretation is receiving much attention and the meteorological Satellite Section is making preparations for a more advanced satellite. The future is speculative, but "a spectrometer is being developed as a possible prototype for a future satellite instrument which may have the capability of measuring the world-wide temperature distribution through the depth of the stratosphere and upper troposphere by measuring the radiation emitted in the 15-micron carbon dioxide band" [3, p. 6]. If complementary measurements on pressure and humidity can be developed by some means, weather practice should be altered toward centralized data gathering.

3. Weather Data Processing of a Routine Nature

3.1 Types of processing

Most weather data processing of an operational character is accommodated by the following outline:

A. Data generation and assembly

- a. analog to digital conversion
- b. data reduction
- c. character and message encoding
- d. message communication
- e. message assembly
- f. message editing

B. File maintenance and data summarization

a. consistency analyses
b. climatological analyses
c. current weather analyses

C. Forecasting analyses

a. subjective predictions
b. objective predictions

D. Presentation of analyses and objective forecasts

a. output formatting
b. digital to analog conversion (where applicable)
c. printing or map drawing (as applicable)

E. Dissemination of forecasts and analyses

a. control of facsimile and message communication
b. personal briefings
c. other

A study of each item in the outline is beyond the scope of this report; discussion must be directed to matters of most relevance to large-scale data processing machinery. Some appreciation for the system is given by the following quotes [12] :

> The Weather Bureau maintains about 300 offices, manned by full-time personnel, at cities and airports throughout the United States and in Puerto Rico and some of the islands of the Pacific Ocean . . . .
>
> In addition, weather reports are made at about 675 other locations by personnel of certain other Government agencies, notably the Federal Aviation Agency, State and municipal groups, airlines, and private citizens.
>
> Also, the Weather Bureau has approximately 12, 700 substations which provide less complex reports and observations to supplement those taken by principal offices. More than one service may be performed at a substation; hence, weather measurements are available daily from about 5, 000 temperature and 11, 400 precipitation observation points. There are also about 4, 000 substations making telegraphic or telephonic reports of current weather data for agricultural services, flood warnings, hurricanes, and severe storm warnings, etc., and nearly 500 display lights and/or flags for storm warnings to shipping interests on waterways.

. . . In cooperation with the Coast Guard, the U. S. Navy, and certain maritime interests, the Weather Bureau maintains ocean weather stations in both the Atlantic and the Pacific and in the Gulf of Mexico . . . The Weather Bureau maintains a national meteorological center at Suitland, Md., which analyzes weather conditions over the northern hemisphere, prepares guidance prognostic material and disseminates it via facsimile and other rapid communication media to both civil and military forecast offices to support their weather service activities . . .

A general public weather service of daily weather bulletins, forecasts, warnings, and advices for agriculture, business, commerce, industry, etc., is provided through the field stations of the Weather Bureau. This information is localized for the area of responsibility assigned to each office, based upon the State forecasts issued from 33 district and State forecast offices. Forecasts are regularly available four times daily to cover weather developments expected during the ensuing 36 to 48 hours. In critical situations warnings of storms and cold waves are issued as developments warrant, without regard to regular forecasting schedules.

Weather bulletins and forecasts are published by nearly all daily newspapers and broadcast by most commercial radio and television stations in the country.

Five-day weather outlook statements are issued for all parts of the United States 3 times weekly. A 30-day outlook is issued at Washington twice a month . . .

The Aviation Weather Service operates 24 hours every day to supply information and weather forecasts for air operations in the United States and over transocean routes. The domestic service consists of about 240 airport stations, connected by teletypewriter circuits, operated by the Federal Aviation Agency, which collect and exchange detailed reports hourly on weather conditions along the civil airways and disseminate forecasts and other current weather information. Pre-flight weather briefing is available at all airport stations. Operational forecasts for aviation are issued every 6 hours by 25 Flight Advisory Weather Service Centers. Flash warnings of potentially hazardous flying weather are issued as needed for safety

of aircraft in flight. Under agreement with foreign governments the Weather Bureau provides forecasts for many overseas air routes. The international air routes are served by specialized forecasting and pilot briefing services, with special attention to transoceanic and high altitude flights. These forecasts are disseminated to international airports and interested subscribers by long-line teletypewriter and facsimile circuits . . .

The Climatological Service collects, summarizes, publishes, and analyzes data from about 11,000 substations manned by unpaid cooperative observers and about 300 regular Weather Bureau stations. Data are included for the United States, Puerto Rico, and Trust Territories. These data are useful to agriculture, commerce, and transportation.

For the data collection program, three Weather Records Centers are operated at San Francisco, Calif., Kansas City, Mo., and Chattanooga, Tenn. The major portion of the analytical work is carried at the National Weather Records Center at Asheville, N. C. This is also the central repository for all American weather records. It also has large collections of foreign and oceanic weather data. Facilities, including electronic computers, are available for large scale data checking and summarizations as well as for the printing of regular monthly and annual summaries . . .

The Army, the Navy, the Air Force, the National Aeronautics and Space Administration (NASA), and a number of miscellaneous agencies generate special requirements for weather services. As general policy, these agencies use as much as possible of the basic weather services of the Weather Bureau and of the communication services of the Federal Aviation Agency. In addition, they carry out operational tasks of their own, and they support the Weather Bureau in efforts to expand and improve service.

The most influential of these agencies has been the Air Force, which has its own Air Weather Service. The operations of the Air Force necessitate terminal and airway forecasts for unusual places, forecasts for missile operations, and weather information for emergency landings and in-flight refueling operations. Naturally the Air Force is acutely interested in sources of weather information that do not black out in time of war.

According to Kolsky [19], the Atomic Energy Commission also does "weather" computations for the purpose of estimating the air pollution and fallout for a scheduled nuclear test. Weather forecasting is not attempted; the procedure is to take upper-air observations at the latest possible time, then to make a rapid computation of the joint implications of the explosive and the prevailing winds.

3.2 Principal computer installations

For climatological analyses at Asheville, N. C., the Air Weather Service has a 705-II; the Weather Bureau has two IBM 650's and two Alwac's. The Weather Bureau has decided to replace the small computers by a Minneapolis-Honeywell 800.

The Joint Numerical Weather Prediction Unit (JNWP) is an inter-service effort supported by the Air Force, Navy, and Weather Bureau. JNWP, located at the National Meteorological Center in Suitland, Md., has an IBM 7090. This computer is used partly for research in automatic analysis and prediction, and partly for operations. Although some maps are optional, the operational processing for a day can require

- current weather analyses (30 maps per day),
- objective predictions (62 maps per day), and
- output formatting (92 maps per day). [11]

Twice-daily forecasts of the height of, and the winds at, the 500 millibar surface for 12, 24, 30, 36, 48, 60, 72 and 96 hours ahead constitute the most basic predictions. Predictions are currently made with a "barotropic" model [1, p. 79].

The Air Weather Service has access to an IBM 7090 at Offutt Air Force Base, Omaha, Neb. Detailed operations are not disclosed, but the general mission is to extend and adapt JNWP output to the needs of the Air Force. The Navy is reported to have assigned its CDC 1604 at Monterey, Calif., to weather work, presumably with a mission comparable to that of the Air Force.

3.3 Output editing and presentation

The traditional way of displaying weather analyses and forecasts is via contour lines on a map. The Z-axis may represent height, pressure, temperature or any of a number of derived quantities. An experienced analyst is said to have required about twenty minutes to draw isolines for a typical map [17].

JNWP recently [6] announced operational use of its off-line "Weather Plotter", a mechanical system for drawing and labelling isolines. The unit

was developed and built by Electronic Associates, Inc., of Long Branch, N. J. It accepts magnetic tape from the IBM 7090, performs a digital to analog conversion, and draws continuous lines with an inked stylus. It labels isolines with a four-digit printer. The preparation of map takes less than three minutes; the maps are unquestionably more accurate than hand-drawn maps. The Air Force (Omaha) and the Navy (Monterey) have ordered Weather Plotters [17].

Maps prepared by JNWP go to over 600 recipients by facsimile.

4. Weather Research and Development

4.1 Types of research

The Weather Bureau seeks to improve its operations in three ways: by extending services, by speeding services, and by improving the quality of forecasts. In practice these objectives are not fully independent, and the most exciting research projects are those which could improve operations in all three ways.

In extending services, the Bureau has instituted special services for hurricanes, tornados, floods, forest fires, and frosts. It is also investigating the possibilities of analyses and forecasts for greater heights in the atmosphere.

The Weather Bureau is participating with the Department of Defense and the Federal Aviation Agency in a research and development program for speeding up aviation weather services and expanding services to aviation "in the field of very short range forecasts of the order of two hours or less. Forecasts of this time period require intensive investigation of small-scale atmospheric phenomena which can be observed routinely only by means of a highly dense network of stations" [3, p. 3]. The impetus for this effort has come largely from the Air Force, which calls its program "Weather Observation and Forecasting System 433-L." This program leans, correctly enough, upon a realization that additional speed in weather processing must come largely from reduction of human intervention in the system. However, the original program statement appears to have underestimated the fundamental and practical problems that lie in the way of automatic short-range forecasting. The 433-L program originally called for a computer and a real-time test network of up to seven weather stations in 1960. United Aircraft, the prime contractor, has not released much information but it is certain that no test network is in operation. The program has dissipated into a study contract, and its future is uncertain.

The Weather Bureau, in cooperation with the National Bureau of Standards, continues development work on an automatic weather observation

station. The latest model, AMOS IV, incorporates a small digital computer for data reduction purposes [15] .

The most far-reaching projects concern satellite meteorology and objective forecasting. A section will be devoted to each.

4.2 Weather satellite experiments

In cooperation with NASA, the Weather Bureau is making plans for TIROS III in 1961, a series of four Nimbus satellites starting in 1962, and a series of five Aeros starting in 1964 [18] . These flights will test various instruments, communication methods, satellite stabilization methods, and orbits. It is already apparent that operational satellites could eliminate the need for hurricane reconnaissance flights, could help detect tornado-spawning conditions, and could provide cloud pictures of great value in time of war. Much remains to be done before the true value of these devices to forecasters will be known.

A command computer will undoubtedly be helpful in communications with the satellites, and there will be substantial data reduction and coordinate rectification tasks, but no tremendous data processing needs are evident in this area.

4.3 Objective forecasting experiments

Such experiments include a variety of small efforts to develop and test forecasting rules and regression formulae. None currently justify data processing requirements at all comparable with those being discussed for dynamic model development.

In the area of dynamic models, work is being done to improve the JNWP model. The recognized approximations in this model are constantly being tested by experience. JNWP has been evolving a model for about five years, with slow but recognized success, and there is no indication that it will, or that it should, suddenly decide to increase the experimental dimension of its efforts.

The JNWP model is dwarfed by an experimental "Mark II" model currently being developed by Smagorinsky. Mark II is described as a "hydrostatic forecast system" in the technical literature [1, p. 54]. Plans are being made for the Weather Bureau to procure time, starting early in 1962, on a Stretch that is scheduled for the National Bureau of Standards. IBM is cooperating with the Weather Bureau in developing a computer program for Mark II. The numerical approach, as planned, is one of finding an explicit solution to a system of non-linear partial differential equations. The method to be used requires the solutions for two time steps to compute

a solution for the third time step.

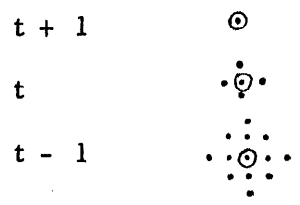
Before formulating Mark II, experience was gained through 704 computations on a Mark I model. (Mark I appears to be roughly comparable to the JNWP model in computation requirements.) Table I, obtained from the Weather Bureau [10], shows that data storage requirements for Mark II are up by a factor of 100 and the multiplication requirements are up by a factor of 600.

Table I: Mark I and II Compared

Statistic	Mark I	Mark II
Points per level	1.3_X10^3	1.5_X10^4
Levels	2	10
Variables per point	2-1/2	4
Total pieces of data (one time period)	6.5_X10^3	6_X10^5
Storage per time step (data for three periods)	2_X10^4	2_X10^6
Multiplications per point	10	16
Time steps per day	72	288
Multiplications per day	4.7_X10^6	2.8_X10^9

Visualize the projection of a hemisphere and a 96 x 96 grid overlay. Let one row in the grid constitute a block of data. A block can have up to 96 points, each point has 10 levels, and each level has 4 words of data. A block can therefore contain about 4000 words.

The computation technique, as planned, requires that to compute a point for time period t + 1, one needs points from period t and t - 1 in accordance with the patterns below (circled points have the same coordinates):



It may be seen that computation of a row for time t + 1 requires three rows from time t and five rows from time t - 1. Once the process has been initiated, two new blocks of data must be read for each one that is generated and written. Storage is required for nine blocks of active data, and buffering will take two more. When other storage requirements are included, the total requirement for core storage comes to at least 50,000 words.

The high speed disk is planned for auxiliary storage. Under current estimates the rate of processing is limited, not by in-out tasks, but by computation.

Some of the basic program for Mark II has already been coded by Data Systems Division. The nature of the program is evident from Table II [9]. The 2311 inner-loop executions take the computation through all levels at one point. Assuming 15,000 points (from Table I), and assuming 2.1 microseconds per average execution, the main computation for one time step takes 73 seconds.

Table II: Frequency of Operations in Mark II

	Type of Instruction					
	Fl. Pt.	VFL*	Branch	Index	Misc.	Total
"RAW" INNER LOOP						
Total storage	259	3	18	18	22	320
% storage	80.9	.9	5.6	5.6	6.9	-
Total execution	2025	3	130	117	36	2311
% execution	87.6	.1	5.6	5.1	1.6	-
BCD TAPE OUTPUT						
Total storage	11	117	71	40	42	281
% storage	3.9	41.5	25.2	14.2	14.9	-
GRID PARAMETERS						
Total storage	145	158	81	60	30	474
% storage	30.6	33.3	17.1	12.6	6.3	-
DISK LOGIC (excluding interpolation)						
Total storage	0	196	128	131	113	568
% storage	0	34.5	22.5	23.0	19.9	-

*Variable field length

Output selection and presentation methods have not been worked out in detail. Selection is to be designed into the model, and is to be based on the behavior of certain integrals. The most sophisticated presentation technique in Mark I was use of a printer for exhibiting contours. This was done by triple spacing the numeric output and interposing two lines of dummy characters for shading. The evaluation process, like the model and the solution method, is really part of the whole experiment.

5. Implications for Computers

5.1 Speculative goal

It is easy enough to imagine an automatic weather system, based on the Weather Bureau and the Federal Aviation Agency, providing a variety of services to other agencies. Automatic observation stations would be controlled by small computers; hemisphere forecasts would be made centrally by a large computer; area and local weather services would be provided by intermediate computers with access to various other computers. Obstacles to such a system arise from factors that are:

- theoretical,
- observational,
- technological, or
- institutional.

The institutional factors include money, incentive, organizational inertia, politics, etc. Principal observational limitations: at present there is no way to effect a big improvement in data from foreign countries. The technological factors include communication and computation capabilities. In this context a theoretical obstacle is one that prevents observations from being numerically transformed into a useful predictive form.

To the extent that this quick survey has seized on relevant information, the theoretical obstacle is thought to dominate the other three. (Although this situation may be due to institutional sins of the past, it is worth noting that the U. S. has consistently led the U. S. S. R. in numerical forecasting achievements [14] .) This does not mean that everything hangs on theoretical developments. It does mean that a proposal of the 433-L type -- which impinges on theory as well as data, technology and institutions -- must take great care to avoid basic theoretical obstacles.

Godske [5] has expressed the widespread hope that a merger of the dynamic model and regression model approaches will eventually permit comprehensive numerical forecasting .

5.2 Machine configurations

The weather complex now has three main uses for a large computer. These are for:

(1) programs of the 433-L type,

(2) operations of the JNWP type, and

(3) experiments of the Mark II model type.

Little has been said in this report of 433-L requirements because of the speculative nature of the program. On the basis of the original statements, Campbell [7] and Campbell and Kolsky [8] determined that a Stretch-like computer was appropriate. Such a machine needs practically everything: fast in-out, sophisticated interrupt, auxiliary storage that is fast and big, editing capabilities, and fast floating-point arithmetic. Chances are that Stretch would be large enough.

JNWP's requirements, if and when they expand, can be met by a Stretch-like computer.

All this can be affected by the outcome of experiments on Mark II, which may lead to larger operational models, to larger experimental models, to smaller operational models, or to nowhere. If history repeats itself, larger experimental models and faster computers will be employed. In this latter case, with emphasis on fast arithmetic, a severe memory fetch and memory bus problem is inevitable. Joint efforts in numerical analysis, program design and machine organization will probably be required to get this problem down to manageable size.

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